

Topic: Factorising Expressions

EQ: How do I factorise expressions?

Factorising is the opposite of distributing (expanding)

$2(3a-6)$
 $\xrightarrow{\text{Factorising is putting into brackets}}$
 $6a-12$
 $\xrightarrow{\text{distributing/expanding (getting rid of the brackets)}}$

Question

$2a + 6$

Working Out

① 2: ^{Factors} 1, 2
6: 1, 2, 3, 6

② $2(\quad)$

③ $2(a + 3)$

Explanation

① List the factors for each number, find the HCF $\rightarrow$ Highest Common Factor

ask yourself
"What is the highest number I can divide both terms by?"

② Put your HCF on the outside of the brackets

③ divide each term by the HCF and put the answer in the brackets

$5a + 15b$

① 5: 1, 5
15: 1, 3, 5, 15

② $5(\quad)$

③ $5a \div 5 = a$ $15b \div 5 = 3b$
 $5(a + 3b)$

*① same $\uparrow$

② HCF is 5 so we put on the outside of the brackets

③ divide each term and put inside the brackets

$2a^2 + 3a$

① 2: ①, 2
3: ①, 3

② $a^2 = a \times a$
 $a = a$

③ $a(\quad)$

④ $2a^2 \div a = 2a$ $3a \div a = 3$

$a(2a + 3)$

① Their HCF is 1 so we can't change the numbers

② The variables are like terms, so we can "pull an a out" $\rightarrow$ divide both by a

③ Put our variable HCF (a) on the outside of the brackets

④ Divide each term and put inside the brackets

Summary

Factorising is when you have the _____ outside the brackets.
We _____ each term inside the bracket by the _____.
Factorising is the _____ of distributing.